

Lecture 8: Proof Notes for Board Work

ISE 5406 – Optimization II, Spring 2026

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Proofs for differentiable convex function characterizations, optimal solution sets, convex maximization at extreme points, and quasiconvex level set characterization.

Green = Steps **Orange** = Key terms **Purple** = Results/Conclusions
Yellow = Write on board *Gray* = Say aloud

Proof Roadmap

Theorem	Key Technique	Est. Time	Priority
First-Order Characterization	Limit argument + convex combination	12–15 min	Full proof
Gradient Monotonicity	Add two first-order inequalities	8–10 min	Full proof
Second-Order Characterization	Taylor expansion + eigenvalue argument	5–8 min	Outline
Optimal Solution Set	First-order condition + gradient constancy	8–10 min	Outline
Extreme Point for Convex Max	Minkowski + Jensen-type inequality	8–10 min	Full proof
Level Set $\Leftrightarrow$ Quasiconvex	Direct application of definitions	5–8 min	Full proof

1 Proof 1: First-Order Characterization of Convexity

Theorem: First-Order Characterization

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be differentiable on S . Then f is **convex if and only if** for any $\bar{\mathbf{x}} \in S$:

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \quad \text{for each } \mathbf{x} \in S.$$

[This is arguably the most important characterization of convex functions. It says the tangent hyperplane is always a global underestimator. Let's prove both directions.]

Direction 1: Convexity $\Rightarrow$ Tangent Underestimator

Step 1: Start from the **definition of convexity**

Assume f is convex. For $\bar{\mathbf{x}}, \mathbf{x} \in S$ and $\lambda \in (0, 1)$:

$$f(\bar{\mathbf{x}} + \lambda(\mathbf{x} - \bar{\mathbf{x}})) \leq (1 - \lambda)f(\bar{\mathbf{x}}) + \lambda f(\mathbf{x})$$

[This is just the convexity definition applied to $\bar{\mathbf{x}}$ and $\mathbf{x}$ with parameter λ .]

Step 2: Rearrange to isolate the **difference quotient**

Rearranging:

$$f(\bar{\mathbf{x}} + \lambda(\mathbf{x} - \bar{\mathbf{x}})) - f(\bar{\mathbf{x}}) \leq \lambda[f(\mathbf{x}) - f(\bar{\mathbf{x}})]$$

Divide by $\lambda > 0$:

$$\frac{f(\bar{\mathbf{x}} + \lambda(\mathbf{x} - \bar{\mathbf{x}})) - f(\bar{\mathbf{x}})}{\lambda} \leq f(\mathbf{x}) - f(\bar{\mathbf{x}})$$

[The left side looks like a difference quotient in the direction $\mathbf{x} - \bar{\mathbf{x}}$.]

Step 3: Take the limit as $\lambda \rightarrow 0^+$

Since f is differentiable:

$$\lim_{\lambda \rightarrow 0^+} \frac{f(\bar{\mathbf{x}} + \lambda(\mathbf{x} - \bar{\mathbf{x}})) - f(\bar{\mathbf{x}})}{\lambda} = \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}})$$

$$\Rightarrow \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq f(\mathbf{x}) - f(\bar{\mathbf{x}})$$

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \quad \checkmark$$

Direction 2: Tangent Underestimator $\Rightarrow$ Convexity

Step 4: Start from the **tangent inequality** applied at a convex combination

Let $\mathbf{x}_1, \mathbf{x}_2 \in S$, $\lambda \in (0, 1)$, and $\mathbf{x}_\lambda = \lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2$.

Apply the tangent inequality from $\mathbf{x}_\lambda$ to $\mathbf{x}_1$ and from $\mathbf{x}_\lambda$ to $\mathbf{x}_2$:

$$f(\mathbf{x}_1) \geq f(\mathbf{x}_\lambda) + \nabla f(\mathbf{x}_\lambda)^T(\mathbf{x}_1 - \mathbf{x}_\lambda)$$

$$f(\mathbf{x}_2) \geq f(\mathbf{x}_\lambda) + \nabla f(\mathbf{x}_\lambda)^T(\mathbf{x}_2 - \mathbf{x}_\lambda)$$

[We have two inequalities. Now we combine them with the right weights.]

Step 5: Multiply and add with **convex combination weights**

Multiply the first by λ and the second by $(1 - \lambda)$:

$$\begin{aligned} \lambda f(\mathbf{x}_1) &\geq \lambda f(\mathbf{x}_\lambda) + \lambda \nabla f(\mathbf{x}_\lambda)^T(\mathbf{x}_1 - \mathbf{x}_\lambda) \\ (1 - \lambda)f(\mathbf{x}_2) &\geq (1 - \lambda)f(\mathbf{x}_\lambda) + (1 - \lambda)\nabla f(\mathbf{x}_\lambda)^T(\mathbf{x}_2 - \mathbf{x}_\lambda) \end{aligned}$$

Adding:

$$\lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2) \geq f(\mathbf{x}_\lambda) + \nabla f(\mathbf{x}_\lambda)^T \underbrace{[\lambda(\mathbf{x}_1 - \mathbf{x}_\lambda) + (1 - \lambda)(\mathbf{x}_2 - \mathbf{x}_\lambda)]}_{=0}$$

[The key observation: the gradient term vanishes because $\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 = \mathbf{x}_\lambda$.]

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) = f(\mathbf{x}_\lambda) \leq \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2) \quad \checkmark$$

Teaching Notes

- **Key Insight:** The forward direction uses a limit argument (“zoom in on the tangent”); the reverse uses a “global to local” combination argument.
- **Board Diagram:** Draw a convex curve, mark $\bar{\mathbf{x}}$, draw the tangent line, show that the curve lies above the tangent everywhere. This is the single most important picture for understanding gradient descent.
- **Time estimate:** 12–15 minutes for both directions.
- **Connection:** Mention that this is WHY gradient descent works: at any point, the gradient tells us a direction of descent, and the first-order condition guarantees we can improve.

Common Student Confusions

1. **“Why do we need S open?”** Because we need differentiability at interior points. If $\bar{\mathbf{x}}$ is on the boundary, the gradient may not be defined. For closed S , we use subgradients (Lecture 5).
2. **“Does this mean $\nabla f(\bar{\mathbf{x}}) = 0$ at the minimum?”** Only for unconstrained minima over open sets. For constrained problems, we need KKT conditions (Lecture 6).
3. **“Why is the gradient term zero in Step 5?”** Because $\lambda(\mathbf{x}_1 - \mathbf{x}_\lambda) + (1 - \lambda)(\mathbf{x}_2 - \mathbf{x}_\lambda) = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 - \mathbf{x}_\lambda = \mathbf{x}_\lambda - \mathbf{x}_\lambda = \mathbf{0}$.

2 Proof 2: Gradient Monotonicity

Theorem: Gradient Monotonicity

Let $S \subseteq \mathbb{R}^n$ be a nonempty open convex set and $f : S \rightarrow \mathbb{R}$ be differentiable. Then **f is convex if and only if:**

$$[\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T (\mathbf{x}_2 - \mathbf{x}_1) \geq 0 \quad \text{for all } \mathbf{x}_1, \mathbf{x}_2 \in S.$$

[This is a beautiful characterization. For scalar functions, it says the derivative is nondecreasing. In general, it says the gradient is a monotone operator.]

Direction 1: Convexity $\Rightarrow$ Gradient Monotonicity

Step 1: Apply the **first-order condition** twice

Since f is convex, the first-order condition gives:

$$\begin{aligned} f(\mathbf{x}_1) &\geq f(\mathbf{x}_2) + \nabla f(\mathbf{x}_2)^T(\mathbf{x}_1 - \mathbf{x}_2) && \text{(from } \mathbf{x}_2 \text{ to } \mathbf{x}_1) \\ f(\mathbf{x}_2) &\geq f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) && \text{(from } \mathbf{x}_1 \text{ to } \mathbf{x}_2) \end{aligned}$$

Step 2: Add the two inequalities

Adding:

$$f(\mathbf{x}_1) + f(\mathbf{x}_2) \geq f(\mathbf{x}_2) + f(\mathbf{x}_1) + \nabla f(\mathbf{x}_2)^T(\mathbf{x}_1 - \mathbf{x}_2) + \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1)$$

Simplify:

$$\begin{aligned} 0 &\geq \nabla f(\mathbf{x}_2)^T(\mathbf{x}_1 - \mathbf{x}_2) + \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1) \\ 0 &\geq -[\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T(\mathbf{x}_2 - \mathbf{x}_1) \end{aligned}$$

$$[\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0 \quad \checkmark$$

Direction 2: Gradient Monotonicity $\Rightarrow$ Convexity

Step 3: Use the **Mean Value Theorem**

For $\mathbf{x}_1, \mathbf{x}_2 \in S$, by MVT there exists $\lambda \in (0, 1)$ such that:

$$f(\mathbf{x}_2) - f(\mathbf{x}_1) = \nabla f(\mathbf{x}_\lambda)^T(\mathbf{x}_2 - \mathbf{x}_1)$$

where $\mathbf{x}_\lambda = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$.

Step 4: Apply **gradient monotonicity** to show the first-order condition

We need to show $f(\mathbf{x}_2) \geq f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1)$.

By MVT: $f(\mathbf{x}_2) - f(\mathbf{x}_1) = \nabla f(\mathbf{x}_\lambda)^T(\mathbf{x}_2 - \mathbf{x}_1)$.

By gradient monotonicity with $\mathbf{x}_\lambda$ and $\mathbf{x}_1$ (note $\mathbf{x}_\lambda - \mathbf{x}_1 = (1 - \lambda)(\mathbf{x}_2 - \mathbf{x}_1)$):

$$\begin{aligned} [\nabla f(\mathbf{x}_\lambda) - \nabla f(\mathbf{x}_1)]^T(\mathbf{x}_\lambda - \mathbf{x}_1) &\geq 0 \\ (1 - \lambda)[\nabla f(\mathbf{x}_\lambda) - \nabla f(\mathbf{x}_1)]^T(\mathbf{x}_2 - \mathbf{x}_1) &\geq 0 \end{aligned}$$

Since $1 - \lambda > 0$: $\nabla f(\mathbf{x}_\lambda)^T(\mathbf{x}_2 - \mathbf{x}_1) \geq \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1)$.

$$\Rightarrow f(\mathbf{x}_2) - f(\mathbf{x}_1) = \nabla f(\mathbf{x}_\lambda)^T(\mathbf{x}_2 - \mathbf{x}_1) \geq \nabla f(\mathbf{x}_1)^T(\mathbf{x}_2 - \mathbf{x}_1)$$

First-order condition holds $\Rightarrow f$ is convex by Proof 1 $\checkmark$

Teaching Notes

- **Key Insight:** Direction 1 is elegant—just add two inequalities. Direction 2 needs the MVT, which students may find less intuitive.
- **Board Diagram:** Draw $f'(x)$ vs. x for a convex function. The slope is nondecreasing $\Rightarrow$ monotone gradient.
- **Time estimate:** 8–10 minutes. Direction 1 is fast; spend more time on Direction 2.
- **Notation:** Point out that the condition $(\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1))^T(\mathbf{x}_2 - \mathbf{x}_1) \geq 0$ is exactly the definition of a **monotone operator** in operator theory.

Common Student Confusions

1. “Is this the same as $\nabla^2 f \succeq 0$?” Related but different. If f is twice differentiable, $\nabla^2 f \succeq 0$ implies gradient monotonicity (take $\mathbf{x}_2 \rightarrow \mathbf{x}_1$ to recover the Hessian condition). Gradient monotonicity is more general—it doesn’t require twice differentiability.
2. “Why does the MVT apply here?” Because S is convex and open, the line segment $[\mathbf{x}_1, \mathbf{x}_2] \subset S$, and f restricted to this segment is differentiable.

3 Proof 3: Convex Maximization at Extreme Points

Theorem: Extreme Point Solution for Convex Max

Let S be a nonempty compact polyhedral set in $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex. Then $\max_{\mathbf{x} \in S} f(\mathbf{x})$ has an optimal solution that is an **extreme point** of S .

[This generalizes the fundamental theorem of LP. The idea is that convex functions “push outward” so the maximum always occurs at a vertex.]

Step 1: Establish that the **maximum is attained**

Since S is compact (closed and bounded) and f is continuous (convex functions on open sets containing S are continuous),

$$\exists \mathbf{x}^* \in S \text{ such that } f(\mathbf{x}^*) = \max_{\mathbf{x} \in S} f(\mathbf{x}).$$

Step 2: Represent $\mathbf{x}^*$ as a **convex combination of extreme points**

Since S is a polytope, by Minkowski’s theorem (representation theorem for polytopes):

$$\mathbf{x}^* = \sum_{i=1}^k \lambda_i \mathbf{x}_i, \quad \lambda_i \geq 0, \quad \sum_{i=1}^k \lambda_i = 1$$

where $\mathbf{x}_1, \dots, \mathbf{x}_k$ are extreme points of S .

[This is the key structural result about polytopes that we proved earlier in the course.]

Step 3: Apply **convexity of f** (Jensen's inequality direction)

By convexity:

$$f(\mathbf{x}^*) = f\left(\sum_{i=1}^k \lambda_i \mathbf{x}_i\right) \leq \sum_{i=1}^k \lambda_i f(\mathbf{x}_i) \leq \max_{i=1,\dots,k} f(\mathbf{x}_i)$$

The last inequality holds because $\sum \lambda_i = 1$ and $\lambda_i \geq 0$.

Step 4: Conclude by **maximality**

We have:

$$f(\mathbf{x}^*) \leq \max_{i=1,\dots,k} f(\mathbf{x}_i)$$

But $f(\mathbf{x}^*)$ is the maximum of f over all of S , and each $\mathbf{x}_i \in S$, so $f(\mathbf{x}_i) \leq f(\mathbf{x}^*)$.

$\Rightarrow f(\mathbf{x}^*) = \max_i f(\mathbf{x}_i) = f(\mathbf{x}_j)$ for some extreme point $\mathbf{x}_j$.

The extreme point $\mathbf{x}_j$ achieves the maximum of f over S . ✓

Teaching Notes

- **Key Insight:** The proof has three ingredients: (1) compactness $\Rightarrow$ max attained, (2) polytope $\Rightarrow$ vertex representation, (3) convexity $\Rightarrow$ Jensen's inequality. Each ingredient is essential.
- **Board Diagram:** Draw a pentagon with contour lines of a convex function. Show that interior points can always be “beaten” by at least one vertex.
- **Time estimate:** 8–10 minutes.
- **Connection to LP:** For linear f , this gives the fundamental theorem of LP. Emphasize that we're generalizing from linear to convex.

Common Student Confusions

1. **“Does this work for minimization too?”** No! The *minimum* of a convex function over a polytope is NOT guaranteed to be at a vertex. Example: $\min x^2$ over $[-1, 1]$ is at $x = 0$ (interior point). The argument fails because the Jensen inequality goes the wrong direction for minimization.
2. **“What if S is not a polytope?”** The result extends to compact convex sets with extreme points (Bauer's maximum principle), but the proof needs Choquet's theorem instead of Minkowski's.
3. **“Does this work for quasiconvex functions?”** Yes, with the additional assumption of continuity. The proof is different (uses level sets) and is a nice exercise.

4 Proof 4: Level Set Characterization of Quasiconvexity

Theorem: Level Set Characterization

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. Then **f is quasiconvex if and only if** the lower level set $S_\alpha = \{\mathbf{x} \in S : f(\mathbf{x}) \leq \alpha\}$ is convex for each $\alpha \in \mathbb{R}$.

[This theorem gives a geometric test for quasiconvexity: just check that all level sets are convex. It's often the easiest way to verify quasiconvexity in practice.]

Direction 1: Quasiconvex $\Rightarrow$ Convex Level Sets

Step 1: Take two points in S_α and apply the **quasiconvexity definition**

Let $\mathbf{x}_1, \mathbf{x}_2 \in S_\alpha$, so $f(\mathbf{x}_1) \leq \alpha$ and $f(\mathbf{x}_2) \leq \alpha$.

For any $\lambda \in (0, 1)$:

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\} \leq \alpha$$

$$\Rightarrow \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S_\alpha$$

S_α is convex. $\checkmark$

[This direction is almost immediate from the definition. The key step is bounding $\max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\} \leq \alpha$.]

Direction 2: Convex Level Sets $\Rightarrow$ Quasiconvex

Step 2: Start with **any two points** and use the appropriate level set

Let $\mathbf{x}_1, \mathbf{x}_2 \in S$ and $\lambda \in (0, 1)$.

Define $\alpha^* = \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$.

Then $\mathbf{x}_1 \in S_{\alpha^*}$ and $\mathbf{x}_2 \in S_{\alpha^*}$ (both have function values $\leq \alpha^*$).

Step 3: Use **convexity of S_{α^*}**

Since S_{α^*} is convex by assumption:

$$\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S_{\alpha^*}$$

By definition of S_{α^*} :

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \alpha^* = \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$$

f is quasiconvex. $\checkmark$

Teaching Notes

- **Key Insight:** Both directions are short and elegant. The trick in Direction 2 is choosing $\alpha^* = \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}$ as the level.
- **Board Diagram:** Draw a quasiconvex function (e.g., a unimodal bump). Show that slicing at any height α gives a convex set (an interval in 1D). Contrast with a non-quasiconvex function (e.g., $\sin(x)$) where level sets are disconnected.
- **Time estimate:** 5–8 minutes for both directions.
- **Practical Note:** In applications, checking level set convexity is often easier than verifying the definition directly.

Common Student Confusions

1. **“Doesn’t convex level sets imply convexity?”** No! Convex level sets imply quasiconvexity, which is strictly weaker than convexity. Example: $f(x) = x^3$ has convex level sets $((-\infty, \alpha^{1/3}])$ but is not convex.
2. **“What about upper level sets?”** Upper level sets characterize *quasiconcavity*: f is quasiconcave iff $\{\mathbf{x} : f(\mathbf{x}) \geq \alpha\}$ is convex for all α .
3. **“Is the converse of ‘every convex function is quasiconvex’ true?”** No. $f(x) = x^3$ is quasiconvex but not convex. However, if f is both quasiconvex *and* quasiconcave, it need not be linear or affine either (e.g., any monotone function on $\mathbb{R}$ is both).

5 Practice Problems for Students

Additional Practice

1. **First-Order Condition:** Show that $f(\mathbf{x}) = \|\mathbf{x}\|^2$ satisfies the first-order condition directly (without using the Hessian).
2. **Gradient Monotonicity:** Prove that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable and f' is nondecreasing, then f is convex. (This is the scalar case of gradient monotonicity.)
3. **Extreme Point Maximization:** Find the maximum of $f(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 1)^2$ over the unit simplex $S = \{\mathbf{x} \in \mathbb{R}_+^2 : x_1 + x_2 \leq 1\}$. Verify it occurs at a vertex.
4. **Quasiconvexity:** Show that $f(x) = \frac{x}{1+x}$ for $x \geq 0$ is quasiconvex. Is it convex?
5. **Strictly Quasiconvex:** Construct a function that is strictly quasiconvex but NOT quasiconvex. (Hint: consider a function that is constant on an interval.)

6 Notation Reference

Notation Used in This Lecture

$\nabla f(\mathbf{x})$	Gradient of f at $\mathbf{x}$
$\nabla^2 f(\mathbf{x})$	Hessian matrix of f at $\mathbf{x}$
$A \succeq 0$	A is positive semidefinite
$A \succ 0$	A is positive definite
$S_\alpha = \{\mathbf{x} \in S : f(\mathbf{x}) \leq \alpha\}$	Lower level set at level α
$\mathbf{x}_{\in S} f(\mathbf{x})$	Set of minimizers of f over S
$\mathbf{x}_\lambda = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$	Convex combination